

Table 1: Model performance metrics for all candidate models

Model	Predictors	Max $\hat{R}$	Min ESS	LOOIC	SE	Δ LOOIC	SE	p_{eff}	$n_{\text{hi-k}}$
baseline	$\delta^2\text{H}_p$	1.004	3474	1336.7	53.0	493.5	54.8	3.6	0
baseline.sp	$\delta^2\text{H}_p$ + GP	1.008	1428	913.9	73.3	70.7	22.9	43.5	3
baseline.env	$\delta^2\text{H}_p$, elev, precip	1.004	1000	1235.7	58.1	392.5	44.7	14.0	0
baseline.env.sp	$\delta^2\text{H}_p$, elev, precip + GP	1.003	1428	860.6	81.5	17.4	7.5	52.8	2
baseline.veg	$\delta^2\text{H}_p$, PFT, C4	1.003	3473	1290.7	51.0	447.5	54.8	8.2	0
baseline.veg.sp	$\delta^2\text{H}_p$, PFT, C4 + GP	1.005	928	885.0	74.6	41.8	20.0	48.5	2
full	$\delta^2\text{H}_p$, PFT, C4, elev, precip	1.006	1129	1217.8	57.5	374.6	44.4	17.7	0
full.sp	$\delta^2\text{H}_p$, PFT, C4, elev, precip + GP	1.004	1581	843.2	80.4	0.0	0.5	55.9	2
full.interact	$\delta^2\text{H}_p$, PFT $\times$ $\delta^2\text{H}_p$, C4 $\times$ $\delta^2\text{H}_p$, elev, precip	1.010	971	1218.2	57.6	375.0	44.3	17.9	0
full.interact.sp	$\delta^2\text{H}_p$, PFT $\times$ $\delta^2\text{H}_p$, C4 $\times$ $\delta^2\text{H}_p$, elev, precip + GP	1.006	1743	843.2	80.3	0.0	0.0	56.0	2
elevation_only.sp	$\delta^2\text{H}_p$, elev + GP	1.005	1484	860.7	81.1	17.5	7.3	52.7	2
elevation_c4.sp	$\delta^2\text{H}_p$, elev, C4 + GP	1.003	1458	862.5	80.7	19.3	7.0	52.6	3
c4_only.sp	$\delta^2\text{H}_p$, C4 + GP	1.007	1703	916.1	72.4	72.9	23.9	41.7	2
elevation_c4.interact.sp	$\delta^2\text{H}_p$, C4 $\times$ $\delta^2\text{H}_p$, elev + GP	1.010	1440	862.7	80.9	19.5	7.1	52.4	2

Note: $\hat{R}$ = Gelman-Rubin convergence diagnostic; ESS = effective sample size; LOOIC = leave-one-out information criterion; SE = standard error; p_{eff} = effective number of parameters; $n_{\text{hi-k}}$ = number of high Pareto- k values; GP = Gaussian process; $\delta^2\text{H}_p = \delta^2\text{H}_{\text{precip}}$; elev = elevation.

Table 2: Global parameter estimates for all models

Model	RMSE		R^2		β_0		β_{OIPC}		λ	GP scale (km)	
baseline	21.1	[21.1, 21.2]	0.703	[0.701, 0.704]	-125.7	[-128.4, -122.9]	0.818	[0.783, 0.855]	8.3	[3.3, 18.2]	-
baseline_sp	15.9	[15.4, 16.3]	0.831	[0.822, 0.842]	-139.4	[-150.9, -128.1]	0.608	[0.511, 0.712]	7.2	[2.9, 15.5]	3937 [2453, 6051]
baseline_env	19.8	[19.6, 19.9]	0.740	[0.735, 0.744]	-125.6	[-128.5, -122.5]	0.833	[0.792, 0.875]	7.5	[3.5, 14.5]	-
baseline_env_sp	15.3	[15.0, 15.6]	0.844	[0.837, 0.850]	-153.4	[-167.6, -140.0]	0.494	[0.361, 0.616]	6.8	[3.0, 13.5]	3402 [2081, 5579]
baseline_veg	20.5	[20.4, 20.6]	0.720	[0.717, 0.721]	-120.1	[-123.9, -116.3]	0.808	[0.766, 0.853]	14.7	[5.2, 37.4]	-
baseline_veg_sp	15.6	[15.0, 16.1]	0.837	[0.827, 0.849]	-135.8	[-147.9, -123.6]	0.565	[0.457, 0.677]	8.3	[3.3, 18.0]	4247 [2680, 6128]
full	19.5	[19.4, 19.7]	0.746	[0.742, 0.750]	-120.5	[-125.0, -115.8]	0.827	[0.779, 0.877]	9.6	[4.3, 19.6]	-
full_sp	15.1	[14.8, 15.5]	0.848	[0.841, 0.854]	-151.4	[-167.5, -136.6]	0.444	[0.305, 0.576]	8.0	[3.4, 16.1]	3784 [2359, 5843]
full_interact	19.5	[19.4, 19.7]	0.746	[0.742, 0.750]	-120.5	[-125.1, -115.8]	0.827	[0.779, 0.875]	9.6	[4.0, 20.2]	-
full_interact_sp	15.1	[14.8, 15.5]	0.848	[0.841, 0.854]	-151.7	[-167.1, -136.9]	0.443	[0.304, 0.575]	7.9	[3.4, 16.1]	3785 [2382, 5873]
elevation_only_sp	15.3	[15.0, 15.6]	0.844	[0.837, 0.850]	-152.8	[-166.9, -139.3]	0.498	[0.365, 0.625]	6.8	[3.0, 13.5]	3412 [2083, 5540]
elevation_c4_sp	15.3	[15.0, 15.6]	0.844	[0.837, 0.849]	-152.2	[-166.6, -138.9]	0.500	[0.367, 0.625]	6.8	[3.0, 13.5]	3421 [2065, 5631]
c4_only_sp	16.0	[15.5, 16.4]	0.830	[0.822, 0.840]	-138.2	[-148.8, -127.7]	0.600	[0.508, 0.699]	7.0	[2.8, 14.8]	3824 [2340, 5939]
elevation_c4_interact_sp	15.3	[15.0, 15.6]	0.843	[0.837, 0.849]	-152.1	[-166.6, -138.9]	0.501	[0.372, 0.625]	6.8	[3.0, 13.3]	3429 [2077, 5651]

Note: Values shown as mean [95% credible interval]. RMSE = root mean squared error; β_0 = global intercept; β_{OIPC} = global slope with respect to $\delta^2\text{H}_{\text{precip}}$; λ = integration length scale; GP scale = Gaussian process length scale in kilometers.

Table 3: Variance decomposition for spatial models

Model	Spatial (%)	Residual (%)	Intercept (%)	Slope (%)
baseline_sp	54.4	45.6	97.1	2.9
baseline_env_sp	66.7	33.3	99.8	0.2
baseline_veg_sp	56.9	43.1	94.0	6.0
full_sp	68.2	31.8	99.7	0.3
full_interact_sp	68.3	31.7	99.7	0.3
elevation_only_sp	66.3	33.7	99.8	0.2
elevation_c4_sp	65.8	34.2	99.8	0.2
c4_only_sp	53.1	46.9	97.7	2.3
elevation_c4_interact_sp	65.7	34.3	99.8	0.2

Note: Variance components shown as percentage of total variance. Spatial = variance explained by spatial Gaussian process; Residual = unexplained variance; Intercept/Slope = proportion of spatial variance attributed to intercept vs. slope components.

Table 4: Environmental covariate coefficients

Model	β_{C4}	β_{tree}	β_{shrub}	β_{grass}	β_{precip}
baseline	-	-	-	-	-
baseline_sp	-	-	-	-	-
baseline_env	-	-	-	-	-0.053 [-0.078, -0.028]
baseline_env_sp	-	-	-	-	0.009 [-0.023, 0.042]
baseline_veg	-0.037 [-0.121, 0.047]	-0.005 [-0.007, -0.003]	0.001 [-0.001, 0.003]	-0.002 [-0.003, -0.001]	-
baseline_veg_sp	0.132 [0.030, 0.232]	-0.001 [-0.003, 0.001]	0.002 [-0.000, 0.004]	-0.002 [-0.004, -0.001]	-
full	-0.069 [-0.160, 0.022]	-0.003 [-0.005, -0.001]	0.001 [-0.001, 0.003]	-0.002 [-0.004, -0.001]	-0.027 [-0.055, 0.002]
full_sp	0.049 [-0.058, 0.157]	-0.002 [-0.004, 0.001]	0.002 [-0.001, 0.004]	-0.002 [-0.003, -0.001]	0.017 [-0.016, 0.050]
full_interact	-0.069 [-0.160, 0.024]	-0.003 [-0.005, -0.001]	0.001 [-0.001, 0.003]	-0.002 [-0.004, -0.001]	-0.027 [-0.055, 0.002]
full_interact_sp	0.049 [-0.057, 0.156]	-0.002 [-0.004, 0.001]	0.002 [-0.001, 0.004]	-0.002 [-0.003, -0.001]	0.017 [-0.017, 0.050]
elevation_only_sp	-	-	-	-	-
elevation_c4_sp	0.032 [-0.066, 0.129]	-	-	-	-
c4_only_sp	0.111 [0.014, 0.203]	-	-	-	-
elevation_c4_interact_sp	0.032 [-0.067, 0.131]	-	-	-	-

Note: Coefficient estimates shown as mean [95% credible interval]. β_{C4} = C4 grass fraction effect; β_{tree} , β_{shrub} , β_{grass} = plant functional type effects; β_{precip} = precipitation effect. Dashes indicate parameters not included in model.